Cloud Computing Architecture Whitepaper
Section 1: AWS EC2 Compute Services

Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity in the Amazon Web Servic

Key EC2 Instance Pricing Models:
1. On-Demand Instances: Pay for compute capacity by the hour or second with no long-term commitments or up
2. Reserved Instances: Provide significant discounts (up to 72%) compared to On-Demand pricing with 1-year or
3. Spot Instances: Allow bidding on spare Amazon EC2 computing capacity for up to 90% discount off the On-De
4. Savings Plans: Flexible pricing model offering lower prices on EC2, Lambda, and Fargate usage in exchange

Section 2: Amazon S3 Storage Services

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, d

Durability and Availability Specifications:
- Data Durability: Amazon S3 Standard, S3 Intelligent-Tiering, and S3 Glacier storage classes are designed for 9
- Availability SLA: Amazon S3 Standard is designed for 99.99% availability of objects over a given year.
- Storage Classes:
  * S3 Standard: For frequently accessed data with low latency and high throughput.
  * S3 Standard-IA: For data accessed less frequently, but requiring rapid access when needed.
  * S3 Glacier Flexible Retrieval: Secure, durable, and low-cost storage for archiving with retrieval times from 1 m
  * S3 Glacier Deep Archive: Lowest-cost storage class where retrieval times range from 12 to 48 hours.

Section 3: Virtual Private Cloud (VPC) and Network Security

Amazon Virtual Private Cloud (Amazon VPC) allows provisioning a logically isolated section of the AWS Cloud w

Network Security Rules:
- Security Groups: Act as a virtual firewall for associated Amazon EC2 instances. Security Groups are stateful; if
- Network Access Control Lists (NACLs): Act as a firewall for controlling traffic in and out of one or more subnets
- Default Limits: By default, you can create up to 5 VPCs per AWS Region, though this quota can be increased b